

Stress Patterns in Intra-word Code-switching

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This pilot study examines the phonology of intra-word switches in Kazakh–Russian, presenting an acoustic analysis from a production experiment.

We find evidence of convergence in word-internal switches, with hybrid patterns emerging from the interaction of stress placement (longer duration, greater intensity) and vowel reduction as a secondary effect of stress.

We also analyze this pattern within an Optimality-Theoretic framework.

Phonology of Intra-word CS

Intra-word code-switching is the phenomenon of combining the elements of different languages within a single word, for example:

English root + Spanish affixes (MacSwan, 2005):

(1) /it-eando/ eat-DUR “eating”
(2) /it-ará/ eat-FUT/3SG “will eat”

How do the phonological systems of two languages interact in intra-word code-switching? Does the phonology of L1 or L2 dominate? Or do hybrid patterns emerge?

- Stefanich et al. (2019) note mixed views on intra-word CS phonology and limited acoustic/experimental data.
- This study focuses on a specific case of Kazakh-Russian code-switching.

Stress in Kazakh vs. Russian

Kazakh: final syllable is prominent (Mukhamedova, 2015)

Acoustic correlates: duration (McCollum & Chen, 2021)

al.má - ‘apple’
al.ma.lár - ‘apple-PL’
al.ma.lar.dó - ‘apple-PI-ACC’

Russian: has lexical stress

(Jouravlev & Lupker, 2014)

duration & intensity & vowel quality (Chrabaszc et al., 2014)

zá.mok ‘castle’ vs. za.mók ‘lock’
stra.ná ‘country’ vs. strá.ny ‘countries’
ko.le.só ‘tire’ vs. ko.lé.sa ‘tires’

In a Kazakh-Russian CS context, Russian nouns with different stress patterns are often inflected with Kazakh suffixes:

(1) pól’za-men benefit-INST ‘with benefit’
(2) stená-da wall-LOC ‘on the wall’
(3) ópít-qa experience-DAT ‘on experience’
(4) seló-ny village-ACC ‘the village’

Does the addition of Kazakh suffixes to Russian stems affect the stress pattern?

Hypotheses

H1: Kazakh dominant. The stress switches to the last syllable (suffix).
H2: Russian dominant. Inserted Russian stems keep the original stress. stress may also emerge.
H3: Hybrid stress. The Russian stress is retained; a Kazakh-style final stress may also emerge.

/ópít/ + /qa/
/stená/ + /da/

[opitqa:]
[stenada:]

[ópetqa]
[stenáda]

[ópétqa:]
[stenáda:]

Production Experiment

Participants. 4 Kazakh–Russian bilinguals (2F, 2M), residing in Irvine, CA; *Kazakh:* native proficiency; *Russian:* understanding 10/10, speaking 9.5/10, reading 9.5/10 (LEAP-Q).

Stimuli. 40 disyllabic nouns (Kazakh & Russian), elicited in **unsuffixed** and **suffixed** forms.

Carrier phrase (Kazakh). Bul matinnin ishinde __ sozı bar.
“This text contains the word __.”

Methods & measures. Syllables and vowels hand-annotated; extracted: syllable duration (ms), intensity (dB), pitch (f0); vowel formants F1 & F2.

Analysis Framework

Model Stress Correlates

Predictors

Stress Placement

Word Form

Word Form

Language Context

Language Context

Dependent Variables

Syllable Duration

Syllable Intensity

Model Vowel Reduction

Predictors

Vowel Stress

Vowel Type

Vowel Type

Language Context

Language Context

Dependent Variables

F1-F2 Dispersion

Stress Correlates in CS: duration and intensity

We found that **Russian stress remains fixed on the root** (Figure 1 & 2), signaled most reliably by *duration* and *intensity*, while **Kazakh-style final lengthening** is also present (Figure 2 & 3), illustrating a nuanced **hybrid prosodic integration** in intra-word CS.

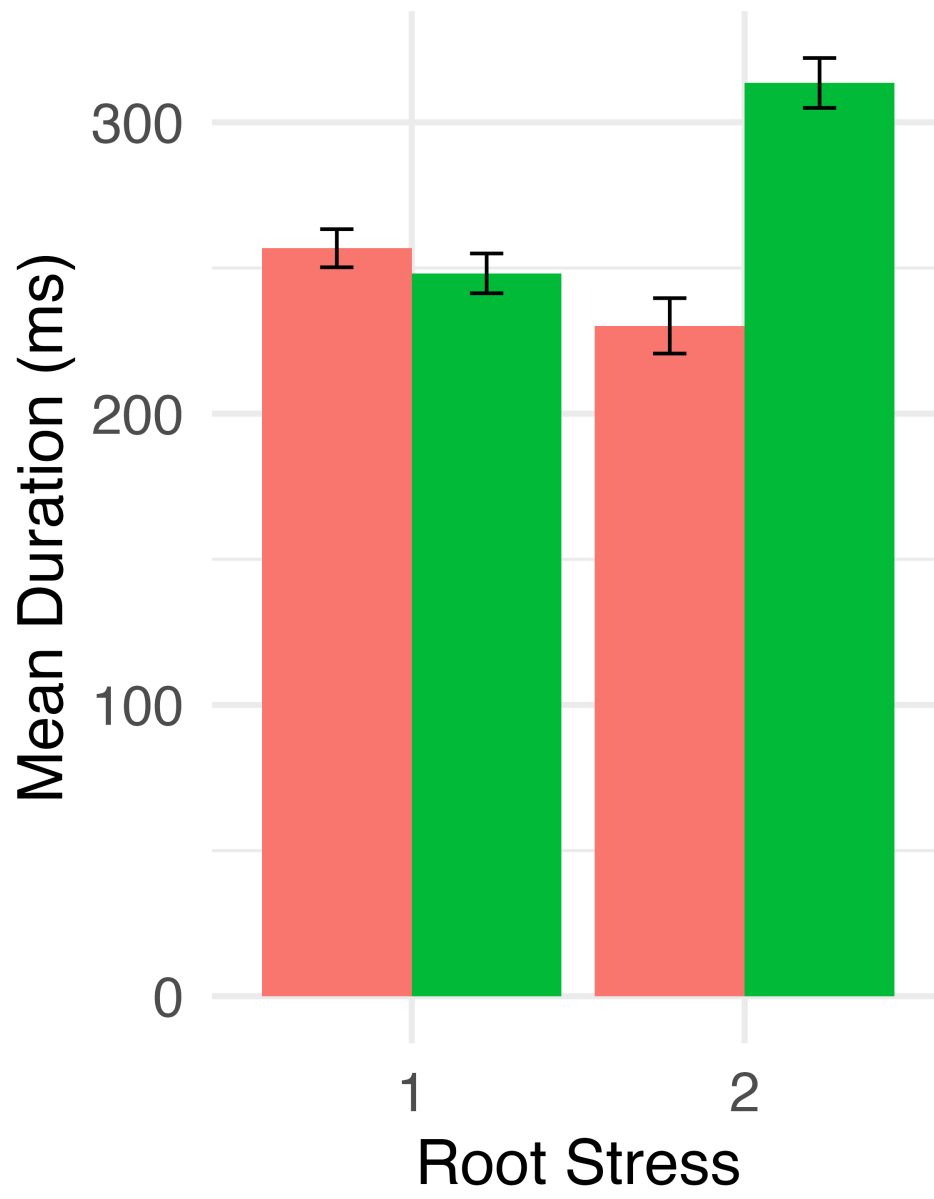


Figure 1: Mean duration in unsuffixed Russian tokens, plotted by root stress position (1 = stress on s1; 2 = stress on s2).

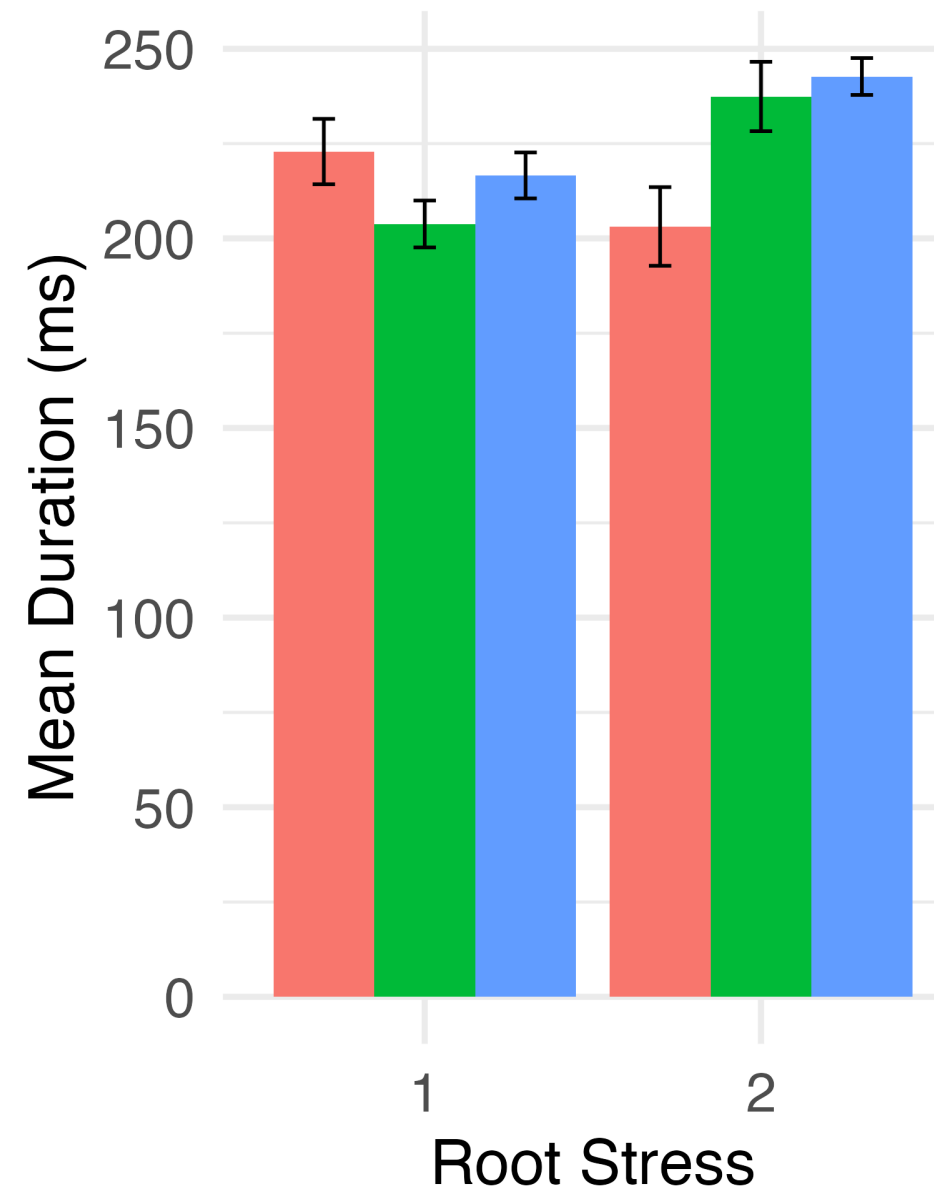


Figure 2: Mean duration in suffixed CS tokens, grouped by root stress position (1 = stress on s1; 2 = stress on s2).

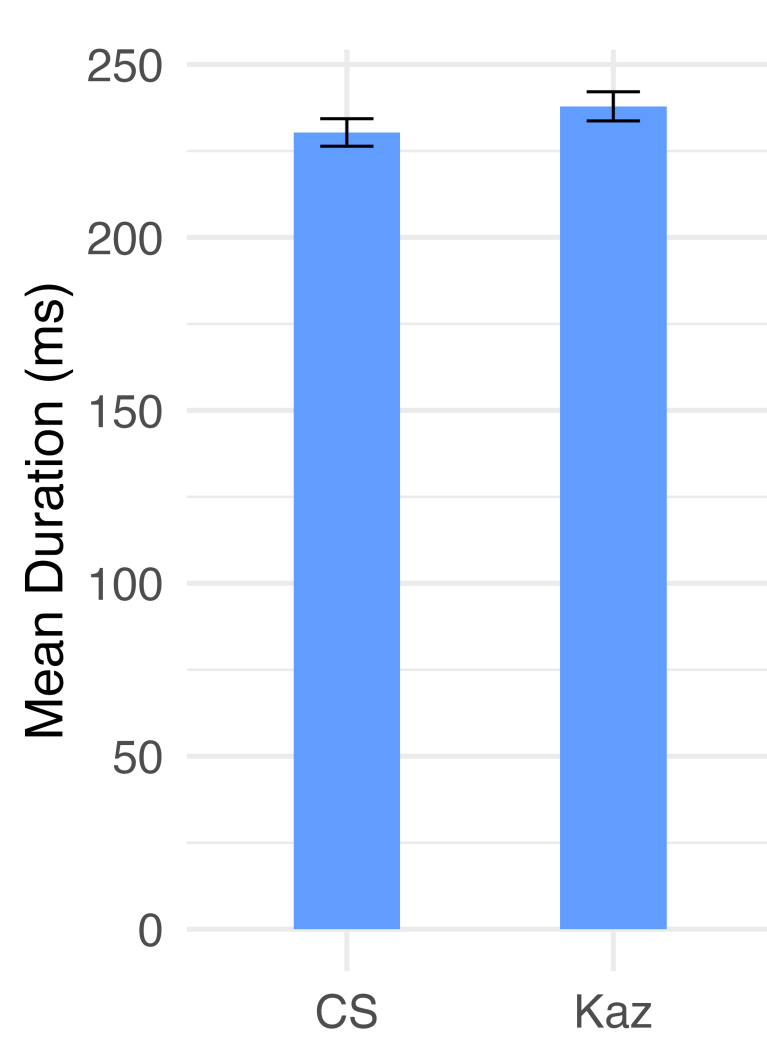


Figure 3: Mean duration of the final syllable (s3) in suffixed Kazakh (non-CS) and CS tokens.

Russian stressed syllables are significantly longer than **unstressed** ones in both unsuffixed and suffixed CS tokens. The same pattern holds for **intensity**: stressed Russian syllables show **higher intensity in both conditions**.

There is **no significant difference** in the **duration** of final syllables (**s3**) in suffixed **non-CS** and **CS** tokens; the same holds for **intensity**.

Vowel Reduction in Russian vs CS

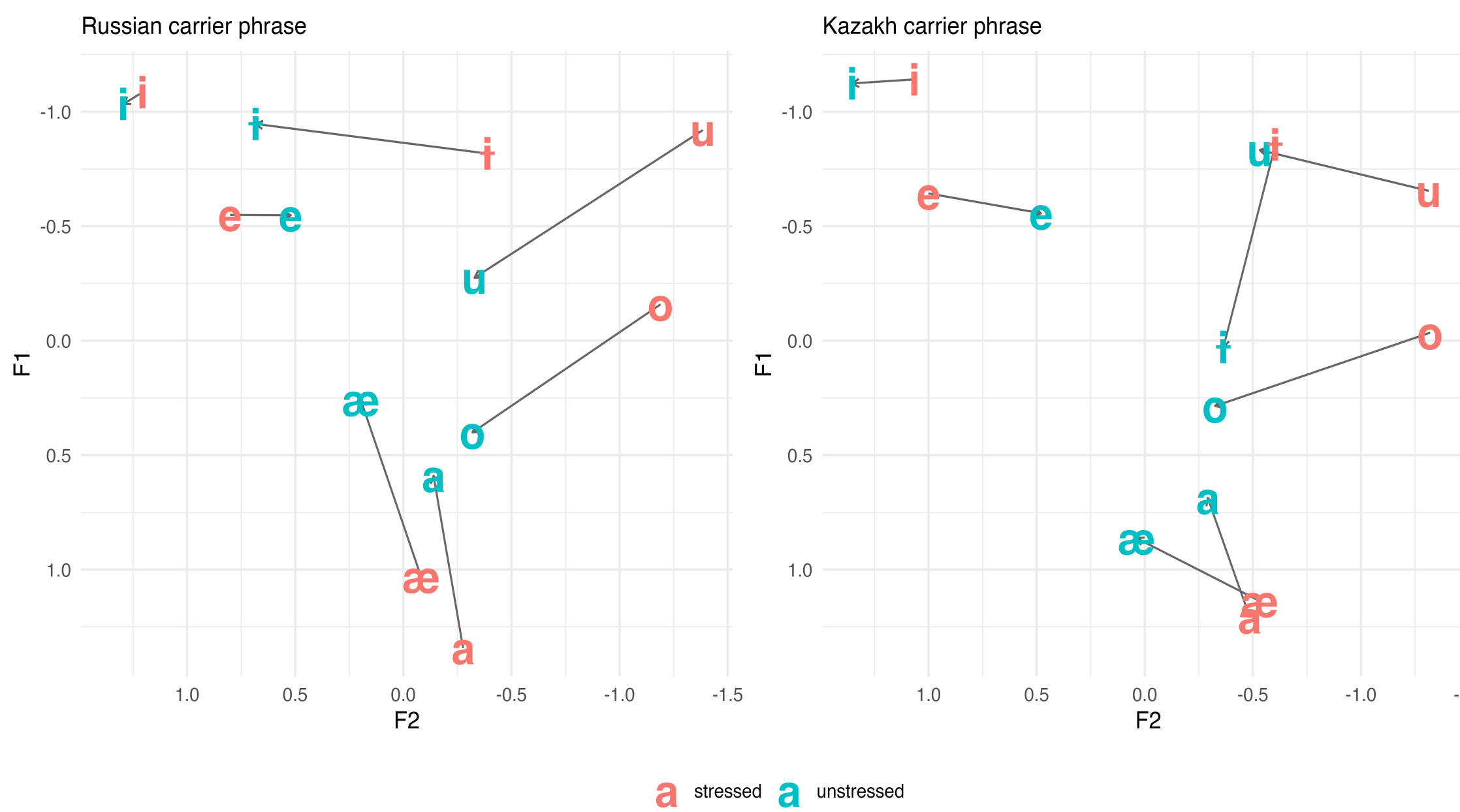


Figure 4: Russian and CS vowel space showing normalized mean F1 and F2 values by vowel category, separated by stress condition.

- Vowel reduction occurs** in both **Russian** and **code-switched CS tokens** (Russian: $\beta = -0.22, p < .001$; CS: $\beta = -0.29, p < .001$).
- Russian stress is characterized by vowel quality: unstressed vowels undergo reduction (Crosswhite, 2000; Padgett & Tabain, 2005).
- Vowel dispersion was measured as *Euclidean distance* from the center of each speaker’s F1–F2 vowel space (Wright et al., 2004).

Context vs Stress

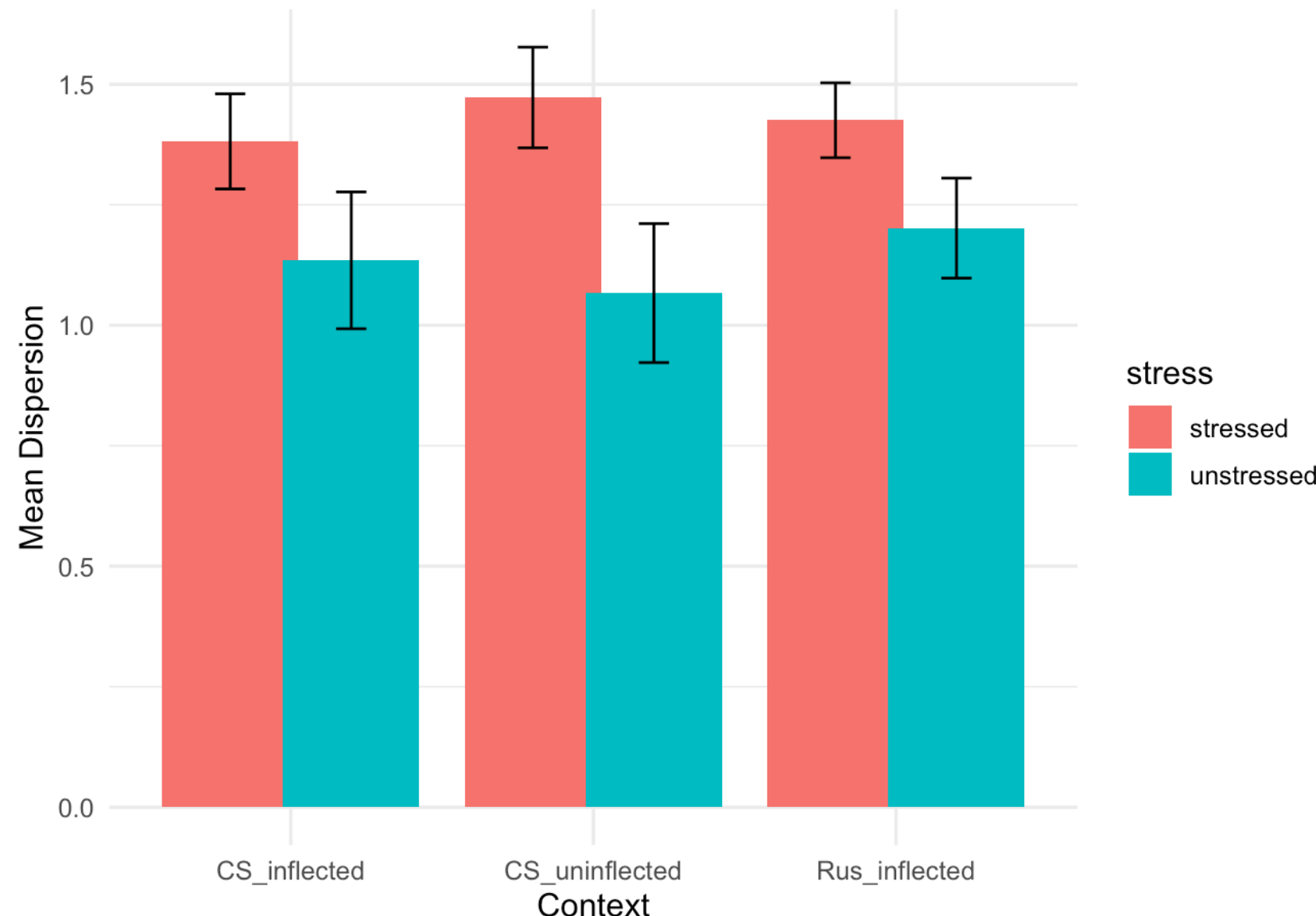


Figure 5: Mean dispersion for stressed and unstressed vowels by context.

- Context** (CS vs. Russian) **did not significantly affect vowel space distance**, suggesting that *phonetic realization is stable* across monolingual (Russian) and mixed CS tokens.
- Significant stress effect:** unstressed vowels had reduced dispersion vs. stressed vowels ($\beta = -0.26, p < .001$).

Phonological Analysis

Stratal OT for Kazakh–Russian Code-Switching (Kiparsky, 2000)

- Constraints:**
ID(Head-σ) preserve stressed syllable; *FullV(root) ban full vowels in unstressed root syllables; ID(root) root faithfulness; AGREE Kazakh vowel harmony; *Struc(Ft) avoid extra feet.
- Cycle 1: Russian stem (/ópít/):**
Ranking: ID(Head-σ) » *FullV(root) » ID(root) » *Struc(Ft)
→ [ópét] (stress preserved; σ2 vowel reduced).
- Cycle 2: Kazakh word-level (/ópet/ + /qA/):**
Ranking: ID(root) » AGREE » *Struc(Ft) » ID(Head-σ)
→ [ópétqa] (root unchanged; suffix harmonizes; no new stress).
- Post-lexical:**
FINAL-LENGTHENING {DUR+} → [ópétqa:]
lengthening only; no phonemic vowel length in Kazakh.

Cycle 1: Russian

/ópít/ ‘experience’	ID(Head-σ)	*FullV(root)	ID(root)	*Struc(Ft)
→ [ópét]			*	*
[ópít]		*!		*
[opít]	*!	*		*
[opít]	*!	**		
[opét]	*!	*	*	

Cycle 2: Kazakh

/ópet/ + /qA/ ‘on experience’	ID(root)	AGREE	*Struc(Ft)	ID(Head-σ)
→ [ópétqa]			*	
→ [opetqa]	*!			*
[ópétke]		*!	*	
[opétqa]	*!		*	*
[opetqá]	*!		*	*

Post-lexical: FINAL-LENGTHENING {DUR+} → [ópétqa:]

Selected References:

Chrabaszc, A., Winn, M., Lin, C.-Y., & Idsardi, W. J. (2014). Acoustic cues to perception of word stress by English, Mandarin, and Russian speakers. *Journal of Speech, Language, and Hearing Research*, 57(4), 1468–1479. Crosswhite, K. M. (2000). Vowel reduction in Russian: a unified account of standard, dialectal, and ‘dissimilative’ patterns. *University of Rochester Working Papers in the Language Sciences*, 1, 107–172. Jouravlev, O., & Lupker, S. J. (2014). Stress consistency and stress regularity effects in Russian. *Language, Cognition and Neuroscience*, 29(5), 605–619. Kiparsky, Paul. 2000. Opacity and cyclicity. In S. Inkelas & K. Hanson (eds.), *The Nature of the Word: Essays in Honor of Paul Kiparsky*, 161–183. MIT Press. MacSwan, J. (2005). Codeswitching and generative grammar: A critique of the MLF model and some remarks on “modified minimalism”. *Bilingualism: Language and Cognition*, 8(1), 1–22. McCollum, A. G., & Chen, S. (2021). Kazakh. *Journal of the International Phonetic Association*, 51(2), 276–298. Mukhamedova, R. (2015). *Kazakh: A comprehensive grammar*. Routledge. Padgett, J., & Tabain, M. (2005). Adaptive dispersion theory and phonological vowel reduction in Russian. *Phonetica*, 62(1), 14–54. Stefanich, S., Cabrelli, J., Hilderman, D., & Archibald, J. (2019). The morphophonology of intra-word codeswitching: Representation and processing. *Frontiers in Communication*, 4, 54. Wright, R., Local, J., Ogden, R., & Temple, R. (2004). Factors of lexical competition in vowel articulation. *Papers in Laboratory Phonology VI*, 75–87.

In intra-word CS,
Russian stress is preserved
while **Kazakh-style final lengthening** applies.